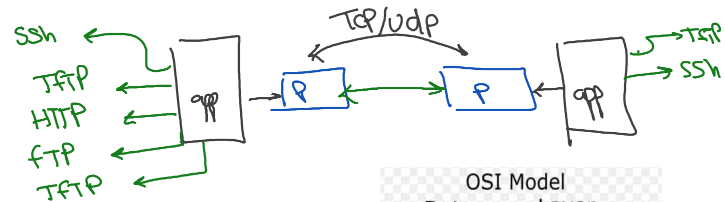
U-boot

Saturday, September 20, 2025 10:14 AM

1. load → Kernel, → dtb } over network
2. rootfs over network
3. extlinux.conf
   → loading of Image



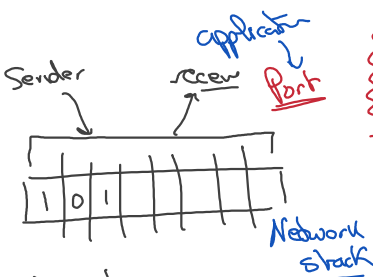
Feature 1

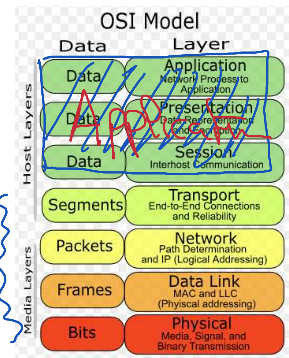
TFTP protocol

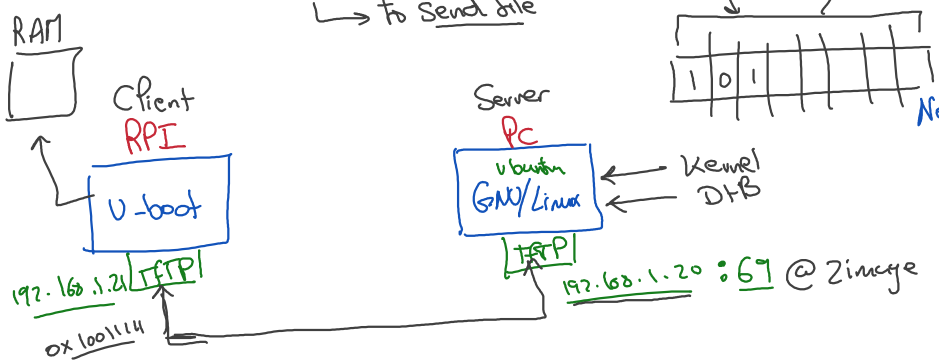
→ application supported

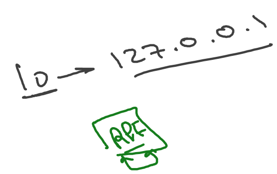
u-boot

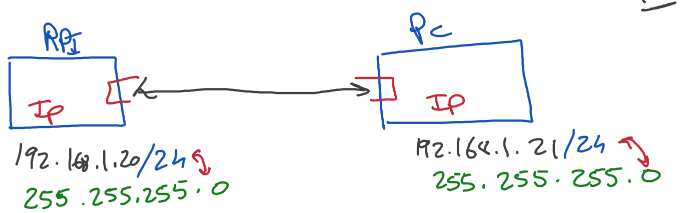
→ to send file

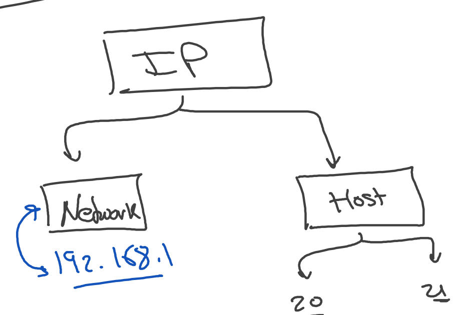












Ubuntu

```
fady@android:/srv/tftp$ cat /etc/default/tftpd-hpa
# /etc/default/tftpd-hpa

TFTP_USERNAME="tftp"
TFTP_DIRECTORY="/srv/tftp"
TFTP_ADDRESS=":69"
TFTP_OPTIONS="--secure --create"
```

rw

Port

① download : sudo apt install TFTPd_hpa

② Configuration will be created by default
↳ /et/default/tftpd-hpa

③ Copy zImage & dtb
↳ /srv/tftp

```
fady@android:~/bootloader$ sudo qemu-system-arm -M vexpress-a9 -kernel u-boot/u-bo
ot --nographic -sd ../NTI/virtualSd.img -net tap -net nic
```

qemu -system -arm

-net tap -net nic

PC
Virtual Interface
tap

qemu
Interface
nic

Internally
PC

Sudo ip a add ____/24 dev tap0

Setenv ipaddr 192.168.1.21

In u-boot

pc IP

tftpboot $kernel_addr_r 192.168.1.20 : zimage
tftpboot $fdt_addr_r 192.168.1.20 : vexpre— .dtb

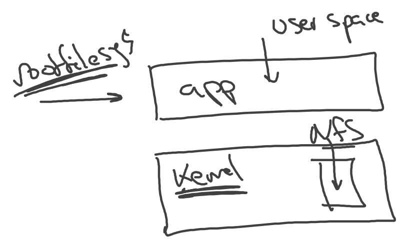
bootz / bootI $kernel_addr_r - $fdt_addr_r

Feature 2

nfs
↳ network file system

↳ bootargs : root= ~~/dev/mmcblk0p2~~
root = /dev/nfs
↳ application (driver)
↳ Network stack

U-boot



Ubuntu

```
# Install an NFS server
sudo apt install nfs-kernel-server

# Add exported directory to `/etc/exports` file, with target ip as follows
/PathToDirectory/rootfs 192.168.0.100(rw,no_root_squash,no_subtree_check)

# Ask NFS server to apply this new configuration (reload this file)
sudo exportfs -r
```

U-boot

```
setenv bootargs console=ttyXXX root=/dev/nfs ip=192.168.0.100:::::eth0 nfsroot=192.168.0.1:/home/fady/Documents/busybox/_install,nfsvers=3,tcp rw
init=/sbin/init
```

PC

nfs, tcp

init pr

rootfs

server

192.168.0.100

API

/home/fady/rootfs

feature 3

Extlinux . conf

① setenv load_kernel load mmc 0:1 $file file

↓ with extlinux

boot
└→ extlinux
  └→ extlinux.conf

Scanflow NTF

FAT mmc 0:1

bootable

```
default Yocto
label Yocto
   kernel /zImage
   ~~fdtdir~~ /
append root=PARTUUID=e170178b-02 rootwait console=ttyS0,115200
```

fdt —— .dtb

bootargs

NTI

fdt_addr_r

boot
↓
zImage

Kernel_add.1 -